

Ashraful Alam

Address: Bijoy 24 Hall, SUST, Sylhet-3100 | Email: ashrafrakib09@gmail.com
Phone: +8801609080302 | LinkedIn: ashraful-alam

RESEARCH INTERESTS

Biostatistics; statistical learning and predictive modeling; machine and deep learning; time-series forecasting; spatial statistics, GIS, and remote sensing; genomic-enviromic data integration; public-health and climate-health data science; infectious disease surveillance; maternal, newborn, and child health in low-resource settings.

EDUCATION

Master of Science (M.Sc.) in Statistics

Shahjalal University of Science and Technology | Sylhet, Bangladesh | May 2025 - Expected Dec 2026

Relevant graduate coursework: Advanced Statistical Inference; Advanced Multivariate Analysis; Time Series Analysis; Econometrics; Advanced Biostatistical Methods; Generalized Linear Mixed Models; Data Mining and Machine Learning.

Bachelor of Science (B.Sc.) in Statistics

Shahjalal University of Science and Technology (SUST) | Sylhet, Bangladesh | Jan 2020 - May 2025

CGPA: 3.82/4.00, with distinction.

Undergraduate Project: Determinants of early childhood mortality in Bangladesh using BDHS survey data 2022

Selected coursework: Probability; Statistical Inference; Regression Analysis; Sampling; Design of Experiments; Statistical Computing; Multivariate Analysis; Stochastic Processes; Biostatistics and Epidemiology; Generalized Linear Models; Research Methodology.

RESEARCH EXPERIENCE

Master's Thesis (ongoing), Department of Statistics, SUST | 2025-Present

- Thesis: Predicting Varietal Performance of Boro Rice Using an Integrated Genomic-Enviromic-Management-Remote Sensing Framework across Diverse Ecosystems of Bangladesh.
- Integrates multi-environment trial, genomic, climate, soil, remote-sensing, and GIS data; planned methods include SpATS, factor-analytic mixed models, reaction norms, genomic prediction, machine/deep learning, location-based validation, uncertainty analysis, and climate-scenario mapping.

Research Assistant, Department of Statistics, SUST | Jul 2024 – Present

Advisor/PI: Professor Dr. Md. Zakir Hossain

- Co-drafted the methodological framework for a major STEM research grant proposal submitted to the UGC Higher Education Acceleration and Transformation (HEAT) Project (2026), designing a community-based Randomized Controlled Trial (RCT) and utilizing Small Area Estimation (SAE) for precision targeting child malnutrition.
- Contribute to faculty-led public-health studies involving maternal-newborn health in haor region of Bangladesh.

Research Associate at Heat Exposure, Health, and Resilience Project | BCSAA-funded project | 2025-2026

Advisor/PI: Professor Dr. Md Jamal Uddin, Department of Statistics, SUST

- Managed and analyzed mixed-methods cross-sectional data from 664 outdoor construction workers across 9 highly vulnerable, extreme-heat-affected urban environments in Bangladesh.
- Implemented a standard Urban Heat Island (UHI) mapping approach utilizing remote sensing of Land Surface Temperature (LST) and land cover classification data.
- Processed geospatial anomalies by computing median clear-sky LST and derived a normalized Heat Island Index to evaluate localized climate-health risks, linking environmental exposure models to public health outcomes (anxiety, depression, productivity loss).
- Contributed to the development and statistical validation of the Outdoor Construction Workers Heat-Resilience Scale.

Research Assistant, Maternal and Newborn Healthcare Services in the Haor Region | Project ID: PS/2024/2/31 | 2024-2026

- Analyzed primary survey data from 800 households across 40 rural clusters in six northeastern Haor districts of Bangladesh.
- Examined determinants of antenatal care, facility delivery, skilled birth attendance, and postnatal care using descriptive statistics and regression models.
- Integrated quantitative results with qualitative findings to support health-service planning recommendations for geographically disadvantaged communities.

Data Analyst and Researcher, Unity Bangladesh Public Health Club | Nov 2024-Feb 2026

- Collaborated on applied public-health research and analyzed survey and surveillance datasets using R, Python, SAS, Stata, and SPSS.

PEER-REVIEWED CONFERENCE AND PUBLICATIONS PROCEEDINGS

- **Alam, A.,** Prodhan, M. E., Hasan, M. N., Lubna, T. S., Islam, M. A., Afroz, M., & Nira, R. A. (2025). **Deep learning-based multivariate time series forecasting of acute respiratory infections among Rohingya refugees in Cox's Bazar, Bangladesh.** In 2025 28th International Conference on Computer and Information Technology (ICCIT). <https://doi.org/10.1109/ICCIT68739.2025.11491162>
- Prodhan, M.E., **Alam, A.,** Hasan, M.N. et al. A comparative study of machine learning algorithms in classifying and predicting exclusive breastfeeding infants in Bangladesh. In: 2024 27th International Conference on Computer and Information Technology (ICCIT); 2024 Dec 19-21; Cox's Bazar, Bangladesh. 2024. p. 2789-93. <https://doi.org/10.1109/ICCIT64611.2024.11022563>
- Hasan, M.N., Prodhan, M.E., **Alam, A.** et al. A comparative study of machine learning algorithms in predicting acute respiratory infection of under-five children in Bangladesh using imbalanced data. In: 2024 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON); 2024; Dhaka, Bangladesh. 2024.p. 274-8. <https://doi.org/10.1109/BECITHCON64160.2024.10962602>
- **Alam, A.,** Prodhan, M.E., Islam, F. et al. The effects of household water, sanitation, and hygiene on early childhood education enrollment: evidence from the 2022 Bangladesh demographic and health survey. *BMC Public Health* 25, 2732 (2025). <https://doi.org/10.1186/s12889-025-23968-7>
- **Alam, A.,** Prodhan, M.E., Akhter, T. *et al.* SWPER-Measured women's empowerment and mental health in Bangladesh: a cross-sectional study from 2022 demographic and health survey. *BMC Women's Health* 26, 119 (2026). <https://doi.org/10.1186/s12905-026-04300-w>

MANUSCRIPTS UNDER REVIEW / IN PREPARATION

- **Alam, A.,** Prodhan, M. E., et al. Learning environment at home and primary school dropout status in Bangladesh: An analysis using Multiple Indicator Cluster Survey (MICS) data. Under review at PLOS ONE.
- Barna, B., **Alam, A.,** Islam, F., Islam, A. A., Modeling under-five mortality using panel econometrics and machine learning: Evidence from 73 countries. Conference abstract accepted for EFAST 2026.
- **Alam, A.,** et al. Prevalence and Determinants of Low Bone Mineral Density Among Adults in Sylhet, Bangladesh. Under review at Scientific Reports
- Himel, M. R., **Alam, A.,** Hasan, M. N., Prodhan, M. E., et al. Heatwave vulnerability among construction workers in heat-prone areas of Bangladesh: A knowledge, attitudes, and practices survey. Manuscript in preparation.
- Himel, M. R., **Alam, A.,** Prodhan, E. et al. Prevalence and Risk Factors for Post-Flash Flood Psychological Morbidity: Insights from the 2024 Bangladesh Flood. Manuscript in preparation

CONFERENCE PRESENTATIONS

- Presenter, "**Deep Learning-Based Multivariate Time Series Forecasting of Acute Respiratory Infections among Rohingya Refugees in Cox's Bazar, Bangladesh,**" 28th International Conference on Computer and Information Technology (ICCIT 2025), Cox's Bazar, Bangladesh, 19-21 December 2025.
- Poster Presenter, "**Meteorological Influences on ARI Burden among Forcibly Displaced Myanmar Nationals in Cox's Bazar: A Multivariate Time-Series Approach.**" International Conference on Applied Statistics and Data Science (ICASDS 2025), Institute of Statistical Research and Training (ISRT), University of Dhaka, Dhaka, Bangladesh. December 28–29, 2025.

- Co-author/Presenter, "Maternal Healthcare Status in the Haor Region of Bangladesh and Factors Influencing Delivery at Health Facilities," 11th INSEARCH International Integrative Research Conference on Governance in Society, Business and Environment, BARD, Cumilla, Bangladesh, 20-21 December 2025.
- Presenter, "Maternal and Healthcare Indicators in the Haor Region of Bangladesh and Determinants of Desired Number of ANC Services," 19th National Statistical Conference, Bangladesh Statistical Association, 22-23 August 2025.

TEACHING, TRAINING, AND MENTORING

Trainer, Hands-on Training on Data Analytics with Python for Economic Research | Department of Economics, Comilla University | 16-17 May 2026

- Delivered hands-on sessions on Python, pandas, data cleaning, descriptive analysis, visualization, OLS regression, diagnostics, panel data, and time-series forecasting.

Lead Instructor, PyNEXT: 7-Day Python Learning Camp | SUST Data Science Club | 2025

- Taught Python programming and data-analysis to students through the SUST Data Science Club.

Academic Mentor in Statistics and Programming | 2022-Present

- Mentor undergraduate and graduate students in Python, statistics, regression analysis, experimental design, time-series analysis, econometric modeling, panel-data methods, and applied machine learning.

LEADERSHIP AND SERVICE

Vice President, SUST Data Science Club | Jan 2025 - Feb 2026

- Organized workshops, seminars, and coding/data-science training activities; mentored peers in statistics, programming, and machine-learning projects.

Executive Committee member, SUST Model United Nations Association | Apr 2021 - Aug 2022

- Worked closely with the Executive Board to plan, manage, and execute the club's activities.

TECHNICAL SKILLS

Statistical methods: Experimental design and Randomized Controlled Trials (RCT); Small Area Estimation (SAE); regression models; Generalized Linear Models; Generalized Linear Mixed Models; linear and factor-analytic mixed models; time-series forecasting; machine learning; deep learning; spatial statistics and spatial epidemiology; genomic-environmental prediction; environmental exposure and panel-data modeling.

Programming, GIS, and software: R, Python, ArcGIS, remote-sensing and raster-data processing (LST and land cover), Git/GitHub, SAS, Stata, SPSS, SQL, Microsoft Office.

AWARDS AND RECOGNITION

- Top 20 finalist, Robi Datathon 3.0, national-level data science competition, 2024.
- Completed B.Sc. in Statistics with distinction, SUST; CGPA 3.82/4.00
- University Scholarship (2019, 2020, 2021), awarded for academic excellence.

LANGUAGES

Bengali: Native | English: Fluent

REFERENCES

Dr. Md Jamal Uddin | Professor, Department of Statistics, SUST
Email: jamal-sta@sust.edu

Dr. Md. Zakir Hossain | Professor, Department of Statistics, SUST
E-mail: mzhossain.bds@gmail.com